

Study of an Interesting q -series from Ramanujan's lost notebooks

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Abstract

We study the family of q -products

$$H_{k,\chi}(q) = \prod_{n=1}^{\infty} (1 - q^n)^{\chi(n) n^{k-2}} = \sum_{m=0}^{\infty} h_{k,\chi}(m) q^m, \quad (k \in \mathbb{Z})$$

where χ is an integer-valued Dirichlet character (in particular, a real quadratic character). This definition is suggested by an identity in Ramanujan's lost notebook, and the definition with the n^{k-2} term is first suggested by [3], but as will become evident, this is a natural definition for us to use.

Our main interest is in the coefficient arithmetic of the series $H_{k,\chi}$ and its normalized coefficients

$$\phi_{k,\chi}(n) := (n!)^{\max(0, 2-k)} h_{k,\chi}(n) \quad (n \geq 0)$$

We prove a uniform factorial integrality theorem for all integers k after this natural normalization (Theorem 3.1), and we record a universal coefficient recursion driven by twisted divisor sums (Proposition 2.1). In the special case $k = 1$ we give a complete Bell-polynomial expression

$$\phi_{1,\chi}(n) = \Phi_n(a_1, a_2, \dots, a_n)$$

(Theorem 4.1), leading quickly to congruence consequences. In particular, we prove the prime-step congruence

$$\phi_{1,\chi}(n+p) \equiv \chi(p) \phi_{1,\chi}(n) \pmod{p}$$

(Theorem 4.2). We also discuss an integral representation for $\log H_{1,\chi}(e^{-t})$ (Section 5), but unsure of correct continuation for a closed form expression so it's left unfinished. We formulate a conjectural strong p -divisibility phenomenon for $k \leq 0$ (Conjecture 6.3). Finally, we include explicit expansions for the conductor 3 character χ_{-3} for $-5 \leq k \leq 12$ to give an idea of what these series look like (Section 7).

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1 Introduction

1.1 Motivation: Ramanujan's product

Ramanujan wrote [1, p. 207] (see also [2]) an identity of the schematic form

$$q^{1/9} \prod_{n=1}^{\infty} (1 - q^n)^{n\chi(n)} = \exp \left(-C - \frac{1}{9} \int_q^1 \frac{f(-t)^9}{f(-t^3)^3} \frac{dt}{t} \right) \quad (1)$$

where χ is a quadratic character and f is a ratio of classical q -products. The constant C is expressible in terms of a Dirichlet L -value; in the case $\chi = \chi_{-3}$ it is also related to the Mahler measure $C = m(1 + x + y)$ [5].

The identity (1) suggests the definition (see [3])

$$H_{k,\chi}(q) := \prod_{n=1}^{\infty} (1 - q^n)^{\chi(n)n^{k-2}} = \sum_{m \geq 0} h_{k,\chi}(m) q^m \quad (2)$$

where $k \in \mathbb{Z}$. Note that Ramanujan's product $\prod (1 - q^n)^{n\chi(n)}$ corresponds to $k = 3$ in (2). The modular/theta-function literature contains many results on generalized eta-functions and Eisenstein series (see, e.g., [4]), and [3] develops applications for (2) in the $k = 2$ case. However, systematic coefficient arithmetic for the broader family (2)—across k and across conductors—appears less developed. The goal here is to record some uniform structural identities and congruence mechanisms for the coefficients.

2 Definitions and basic identities

2.1 Characters and conductor

In this paper we primarily consider real quadratic characters. Equivalently,

$$\chi(n) \in \{0, \pm 1\} \quad (n \in \mathbb{Z})$$

A standard family is given by Kronecker symbols $\chi_D(n) = \left(\frac{D}{n}\right)$. Throughout, $\text{cond}(\chi)$ denotes the conductor of χ .

2.2 Definition of $H_{k,\chi}$ for all $k \in \mathbb{Z}$

Fix $k \in \mathbb{Z}$ and a Dirichlet character χ . For $|q| < 1$ we define

$$H_{k,\chi}(q) := \prod_{n \geq 1} (1 - q^n)^{\chi(n) n^{k-2}} \quad (3)$$

When $k \geq 2$ and χ is integer-valued (real quadratic), the exponents $\chi(n) n^{k-2}$ are integers, so (3) is an ordinary Euler product with coefficients in $\mathbb{Z}[[q]]$.

For $k \leq 1$, the exponents are generally non-integral and we define (3) via the principal logarithm. Let Log denote the principal branch on $\mathbb{C} \setminus (-\infty, 0]$. For $|q| < 1$ we set

$$H_{k,\chi}(q) := \exp \left(\sum_{n \geq 1} \chi(n) n^{k-2} \text{Log}(1 - q^n) \right) \quad (4)$$

which is well-defined and holomorphic for $|q| < 1$ by absolute convergence.

Writing

$$H_{k,\chi}(q) = \sum_{m \geq 0} h_{k,\chi}(m) q^m,$$

the coefficients $h_{k,\chi}(m)$ are rational for real quadratic χ . We introduce a uniform factorial normalization that clears denominators in Theorem 3.1.

2.3 Twisted divisor sums and the logarithmic derivative

Define the twisted divisor sums

$$\sigma_{k,\chi}(m) := \sum_{d|m} \chi(d) d^k.$$

Differentiating (4) and using $\frac{d}{dq} \text{Log}(1 - q^n) = -\frac{nq^{n-1}}{1-q^n}$ yields

$$q \frac{d}{dq} \log H_{k,\chi}(q) = - \sum_{m \geq 1} \sigma_{k-1,\chi}(m) q^m \quad (5)$$

Proof. For $|q| < 1$ we may differentiate termwise

$$\log H_{k,\chi}(q) = \sum_{n \geq 1} \chi(n) n^{k-2} \text{Log}(1 - q^n)$$

since the series converges absolutely and uniformly on compact subsets of the unit disk, differentiating gives

$$\frac{d}{dq} \log H_{k,\chi}(q) = \sum_{n \geq 1} \chi(n) n^{k-2} \frac{d}{dq} \text{Log}(1 - q^n) = \sum_{n \geq 1} \chi(n) n^{k-2} \left(-\frac{nq^{n-1}}{1 - q^n} \right)$$

hence

$$q \frac{d}{dq} \log H_{k,\chi}(q) = - \sum_{n \geq 1} \chi(n) n^{k-1} \frac{q^n}{1 - q^n}$$

Using the geometric series $\frac{q^n}{1 - q^n} = \sum_{r \geq 1} q^{nr}$ (valid for $|q| < 1$), we obtain

$$q \frac{d}{dq} \log H_{k,\chi}(q) = - \sum_{n \geq 1} \chi(n) n^{k-1} \sum_{r \geq 1} q^{nr} = - \sum_{m \geq 1} \left(\sum_{n|m} \chi(n) n^{k-1} \right) q^m$$

Recognizing the inner sum as the twisted divisor sum $\sigma_{k-1,\chi}(m) = \sum_{d|m} \chi(d) d^{k-1}$ yields

$$q \frac{d}{dq} \log H_{k,\chi}(q) = - \sum_{m \geq 1} \sigma_{k-1,\chi}(m) q^m$$

as claimed. $\square$

2.4 A nice recursion on coefficients

Equating coefficients in

$$qH'_{k,\chi}(q) = H_{k,\chi}(q) \cdot q \frac{d}{dq} \log H_{k,\chi}(q)$$

and using (5) gives,

Proposition 2.1 (Universal recursion). *Let $H_{k,\chi}(q) = \sum_{m \geq 0} h_{k,\chi}(m)q^m$ with $h_{k,\chi}(0) = 1$. Then for $m \geq 1$ we have,*

$$h_{k,\chi}(m) = -\frac{1}{m} \sum_{j=1}^m \sigma_{k-1,\chi}(j) h_{k,\chi}(m-j) \quad (6)$$

3 Integrality for all integer k

3.1 Normalization

Let $k \in \mathbb{Z}$ and let χ be an integer-valued Dirichlet character, i.e. $\chi(n) \in \{0, \pm 1\}$ for all n (in particular, any real quadratic character). Define

$$s = \max(0, 2 - k) \quad (7)$$

and set the normalized coefficients

$$\phi_{k,\chi}(n) := (n!)^s h_{k,\chi}(n) \quad (n \geq 0) \quad (8)$$

together with the corresponding integral q -series

$$f_{k,\chi}(q) := \sum_{n \geq 0} \phi_{k,\chi}(n) q^n \in \mathbb{Z}[[q]] \quad (9)$$

3.2 Factorial integrality

Theorem 3.1 (Factorial integrality). *If χ is integer-valued, then for every integer k and every $n \geq 0$ one has*

$$\phi_{k,\chi}(n) \in \mathbb{Z}$$

Equivalently,

$$(n!)^{\max(0, 2-k)} h_{k,\chi}(n) \in \mathbb{Z} \quad (n \geq 0)$$

Proof. Splitting this up as two cases:

$k \geq 2$ Then $\chi(n)n^{k-2} \in \mathbb{Z}$ for every $n \geq 1$. Each factor $(1 - q^n)^{\chi(n)n^{k-2}}$ has integer coefficients, hence $h_{k,\chi}(n) \in \mathbb{Z}$ for all $n \geq 0$. Here $s = \max(0, 2 - k) = 0$, so $\phi_{k,\chi}(n) = h_{k,\chi}(n) \in \mathbb{Z}$.

$k \leq 1$ Write $s = 2 - k \geq 1$ and define $H := H_{k,\chi}$. Using $\log(1 - q^n) = -\sum_{r \geq 1} q^{nr}/r$ and collecting the coefficient of q^m gives

$$\log H(q) = - \sum_{m \geq 1} \frac{1}{m} \left(\sum_{d|m} \chi(d) d^{k-1} \right) q^m \quad (10)$$

Since $k - 1 = 1 - s$, we have $d^{k-1} = d^{1-s} = 1/d^{s-1}$. Define

$$u(m) := \sum_{d|m} \chi(d) \left(\frac{m}{d} \right)^{s-1} \in \mathbb{Z} \quad (11)$$

Then

$$\sum_{d|m} \chi(d) d^{1-s} = \frac{u(m)}{m^{s-1}}$$

and (10) becomes

$$\log H(q) = - \sum_{m \geq 1} \frac{u(m)}{m^s} q^m \quad (12)$$

The exponential formula yields, for $n \geq 1$,

$$h_{k,\chi}(n) = \sum_{\substack{k_1+2k_2+\dots+nk_n=n \\ k_m \geq 0}} \prod_{m=1}^n \frac{(-u(m))^{k_m}}{m^{sk_m} k_m!} \quad (13)$$

Multiplying by $(n!)^s$ gives a sum of terms of the form

$$(n!)^s \prod_{m=1}^n \frac{u(m)^{k_m}}{m^{sk_m} k_m!}$$

For any tuple $(k_1, \dots, k_n)$ with $\sum m k_m = n$, we have that

$$\frac{n!}{\prod_{m=1}^n m^{k_m} k_m!} \in \mathbb{Z} \quad (14)$$

which is the number of permutations of n letters with cycle type $1^{k_1} 2^{k_2} \dots n^{k_n}$. Raising (14) to the s th power and rearranging yields

$$\frac{(n!)^s}{\prod_{m=1}^n m^{sk_m} k_m!} = \left(\frac{n!}{\prod_{m=1}^n m^{k_m} k_m!} \right)^s \cdot \prod_{m=1}^n (k_m!)^{s-1} \in \mathbb{Z}$$

Since $u(m) \in \mathbb{Z}$ by (11), every term contributing to $(n!)^s h_{k,\chi}(n)$ is an integer, hence $\phi_{k,\chi}(n) \in \mathbb{Z}$ for all $n \geq 0$. $\square$

Remark 3.2. The exponent $s = \max(0, 2 - k)$ yields a clean integrality statement for all integer k , but it is not necessarily optimal. Determining the minimal denominators for $h_{k,\chi}(n)$ seems like something interesting to try and prove next.

Remark 3.3. The proof is an application of the exponential formula. Equivalently, $\phi_{k,\chi}(n)$ may be expressed as a complete Bell polynomial in the derivatives of $\log H_{k,\chi}$ at $q = 0$. The integrality ultimately comes from the cycle index of the symmetric group, namely (14).

4 $k = 1$, Bell polynomials and congruences

4.1

For $k = 1$ we write

$$H_{1,\chi}(q) = \prod_{n \geq 1} (1 - q^n)^{\chi(n)/n} = \sum_{n \geq 0} h_{1,\chi}(n) q^n$$

and

$$\phi_{1,\chi}(n) := n! h_{1,\chi}(n) \in \mathbb{Z}$$

Define the divisor-sum sequence

$$b_\chi(m) := \sum_{d|m} \chi(d) \in \mathbb{Z} \quad (15)$$

4.2 Bell polynomial representation

Theorem 4.1. *One has*

$$\log H_{1,\chi}(q) = - \sum_{m \geq 1} \frac{b_\chi(m)}{m} q^m \quad (16)$$

Let

$$a_m := \frac{d^m}{dq^m} \log H_{1,\chi}(q) \Big|_{q=0} = -(m-1)! b_\chi(m) \in \mathbb{Z} \quad (17)$$

Then for every $n \geq 0$,

$$\phi_{1,\chi}(n) = n! h_{1,\chi}(n) = \Phi_n(a_1, a_2, \dots, a_n) \quad (18)$$

where Φ_n is the complete Bell polynomial.

Proof. Expand $\log(1 - q^n) = - \sum_{r \geq 1} q^{nr} / r$ then

$$\log H_{1,\chi}(q) = \sum_{n \geq 1} \frac{\chi(n)}{n} \log(1 - q^n) = - \sum_{m \geq 1} \frac{1}{m} \sum_{d|m} \chi(d) q^m = - \sum_{m \geq 1} \frac{b_\chi(m)}{m} q^m$$

This gives (16). Differentiating termwise gives (17). Finally, the exponential formula for $H_{1,\chi} = \exp(\log H_{1,\chi})$ gives (18). $\square$

4.3 A basic mod 2 observation

An immediate consequence is that since $(m-1)!$ is even for $m \geq 2$, we have $a_m \equiv 0 \pmod{2}$ for $m \geq 2$. Therefore, reducing (18) mod 2 leaves only the a_1^n term:

$$\phi_{1,\chi}(n) \equiv a_1^n \pmod{2}$$

If $\chi(1) = 1$, then $b_\chi(1) = 1$ and $a_1 = -1$, hence $\phi_{1,\chi}(n) \equiv 1 \pmod{2}$ for all $n \geq 0$. (See also [6] for classical congruences involving Bell polynomials.)

4.4 A nice congruence

Theorem 4.2 (Prime-step congruence mod p). *Let χ be an integer-valued Dirichlet character (so $\chi(n) \in \{0, \pm 1\}$), then for every prime p and every $n \geq 0$,*

$$\phi_{1,\chi}(n+p) \equiv \chi(p) \phi_{1,\chi}(n) \pmod{p} \quad (19)$$

In particular, if $p \mid \text{cond}(\chi)$ (so $\chi(p) = 0$), then

$$\phi_{1,\chi}(n+p) \equiv 0 \pmod{p}$$

for all $n \geq 0$.

Proof. Recall from Theorem 4.1 that

$$\log H_{1,\chi}(q) = - \sum_{m \geq 1} \frac{b_\chi(m)}{m} q^m$$

$$b_\chi(m) := \sum_{d|m} \chi(d) \in \mathbb{Z}$$

and set

$$a_m := \frac{d^m}{dq^m} \log H_{1,\chi}(q) \Big|_{q=0} = -(m-1)! b_\chi(m) \in \mathbb{Z}$$

Since $H_{1,\chi}(q) = \exp(\log H_{1,\chi}(q))$, the complete Bell recursion gives, for $n \geq 1$,

$$\phi_{1,\chi}(n) = \sum_{j=1}^n \binom{n-1}{j-1} a_j \phi_{1,\chi}(n-j) \quad (20)$$

for $\phi_{1,\chi}(0) = 1$.

Fix a prime p . Since $(j-1)! \equiv 0 \pmod{p}$ for $j \geq p+1$, we have

$$a_j \equiv 0 \pmod{p} \quad (j \geq p+1) \quad (21)$$

and thus (20) reduces modulo p to

$$\phi_{1,\chi}(n) \equiv \sum_{j=1}^{\min(n,p)} \binom{n-1}{j-1} a_j \phi_{1,\chi}(n-j) \pmod{p} \quad (22)$$

For $1 \leq j \leq p$ we claim

$$\binom{n+p-1}{j-1} \equiv \binom{n-1}{j-1} \pmod{p} \quad (23)$$

Indeed, in $\mathbb{F}_p[x]$ one has $(1+x)^p \equiv 1+x^p$, hence

$$(1+x)^{n+p-1} = (1+x)^{n-1}(1+x)^p \equiv (1+x)^{n-1}(1+x^p) \pmod{p}$$

Comparing coefficients of x^{j-1} for $j-1 < p$ yields (23).

Let $\lambda := \phi_{1,\chi}(p) \pmod{p}$. Using the complete Bell polynomial expansion $\phi_{1,\chi}(p) = \Phi_p(a_1, \dots, a_p)$, all terms are divisible by p except the two partitions $p = 1 + \dots + 1$ and $p = p$, giving

$$\phi_{1,\chi}(p) \equiv a_1^p + a_p \pmod{p} \quad (24)$$

Now $a_1 = -(1-1)!b_\chi(1) = -1$, so $a_1^p \equiv -1 \pmod{p}$. Moreover,

$$a_p = -(p-1)!b_\chi(p) \equiv b_\chi(p) \pmod{p}$$

by Wilson's theorem $(p-1)! \equiv -1 \pmod{p}$. Since the divisors of p are 1 and p ,

$$b_\chi(p) = \chi(1) + \chi(p) = 1 + \chi(p)$$

and therefore (24) yields

$$\phi_{1,\chi}(p) \equiv -1 + (1 + \chi(p)) \equiv \chi(p) \pmod{p} \quad (25)$$

Thus $\lambda \equiv \chi(p) \pmod{p}$.

Proof. A quick proof for (24): by the complete Bell polynomial expansion,

$$\phi_{1,\chi}(p) = \Phi_p(a_1, \dots, a_p) = \sum_{\substack{k_1+2k_2+\dots+pk_p=p \\ k_j \geq 0}} \frac{p!}{\prod_{j=1}^p (j!)^{k_j} k_j!} \prod_{j=1}^p a_j^{k_j}$$

For prime p , the coefficient

$$C(k) := \frac{p!}{\prod_{j=1}^p (j!)^{k_j} k_j!}$$

is divisible by p for every solution $(k_1, \dots, k_p)$ except $k_1 = p$ and $k_p = 1$. indeed $v_p(p!) = 1$, while if $k_p = 0$ then $p \nmid (j!)^{k_j}$ for $j < p$ and also $k_j < p$ so $p \nmid k_j!$, hence $p \mid C(k)$. The two exceptional tuples give $C = 1$ and monomials a_1^p and a_p , respectively. Therefore $\phi_{1,\chi}(p) \equiv a_1^p + a_p \pmod{p}$. $\square$

We prove $\phi_{1,\chi}(n+p) \equiv \lambda \phi_{1,\chi}(n) \pmod{p}$ for all $n \geq 0$. The case $n = 0$ is immediate from the definition of λ . Assume the claim holds for all smaller indices $< n$. Applying (22) with n replaced by $n+p$ and using (21), we obtain

$$\phi_{1,\chi}(n+p) \equiv \sum_{j=1}^p \binom{n+p-1}{j-1} a_j \phi_{1,\chi}(n+p-j) \pmod{p}$$

Using (23), this becomes

$$\phi_{1,\chi}(n+p) \equiv \sum_{j=1}^p \binom{n-1}{j-1} a_j \phi_{1,\chi}((n-j)+p) \pmod{p}$$

Since $n-j < n$, the inductive hypothesis gives $\phi_{1,\chi}((n-j)+p) \equiv \lambda \phi_{1,\chi}(n-j) \pmod{p}$, hence

$$\phi_{1,\chi}(n+p) \equiv \lambda \sum_{j=1}^p \binom{n-1}{j-1} a_j \phi_{1,\chi}(n-j) \pmod{p}$$

By (22), the sum is congruent to $\phi_{1,\chi}(n)$ modulo p . Therefore $\phi_{1,\chi}(n+p) \equiv \lambda \phi_{1,\chi}(n) \pmod{p}$ for all $n \geq 0$. Combining with (25) yields (19). $\square$

Remark 4.3. If $p \nmid \text{cond}(\chi)$ then $\chi(p) = \pm 1$, and iterating (19) gives $\phi_{1,\chi}(n+mp) \equiv \chi(p)^m \phi_{1,\chi}(n) \pmod{p}$, i.e. a period dividing $2p$. If $p \mid \text{cond}(\chi)$, then $\chi(p) = 0$ and $\phi_{1,\chi}(n) \equiv 0 \pmod{p}$ for all $n \geq p$.

5 A potentially interesting local expansion for $k = 1$

Theorem 5.1. *Let χ be a nonprincipal Dirichlet character. For $|q| < 1$ define*

$$H_{1,\chi}(q) := \exp \left(\sum_{n \geq 1} \frac{\chi(n)}{n} \text{Log}(1 - q^n) \right) = \prod_{n \geq 1} (1 - q^n)^{\chi(n)/n}$$

where Log denotes the principal branch on $\mathbb{C} \setminus (-\infty, 0]$. Set $q = e^{-t}$ with $t > 0$, and define

$$b_\chi(m) := \sum_{d|m} \chi(d)$$

as before.

Using $\text{Log}(1 - z) = -\sum_{r \geq 1} z^r/r$ for $|z| < 1$, we have

$$\log H_{1,\chi}(q) = \sum_{n \geq 1} \frac{\chi(n)}{n} \text{Log}(1 - q^n) = - \sum_{n \geq 1} \sum_{r \geq 1} \frac{\chi(n)}{nr} q^{nr}$$

Collecting the coefficient of q^m (write $m = nr$) yields

$$\log H_{1,\chi}(q) = - \sum_{m \geq 1} \frac{b_\chi(m)}{m} q^m \tag{26}$$

$$b_\chi(m) = \sum_{d|m} \chi(d)$$

as before, once again, but now set $q = e^{-t}$. For any $c > 0$,

$$e^{-mt} = \frac{1}{2\pi i} \int_{(c)} \Gamma(s) (mt)^{-s} ds \tag{27}$$

where $\int_{(c)}$ denotes integration along the vertical line $\Re(s) = c$. Insert (27) into:

$$-\log H_{1,\chi}(e^{-t}) = \sum_{m \geq 1} \frac{b_\chi(m)}{m} \cdot \frac{1}{2\pi i} \int_{(c)} \Gamma(s) (mt)^{-s} ds$$

We may interchange sum and integral for $c > 0$ by absolute convergence: indeed $|b_\chi(m)| \leq d(m)$ and $\sum_{m \geq 1} d(m)m^{-1-c} < \infty$. Thus

$$-\log H_{1,\chi}(e^{-t}) = \frac{1}{2\pi i} \int_{(c)} \Gamma(s) t^{-s} \left(\sum_{m \geq 1} \frac{b_\chi(m)}{m^{1+s}} \right) ds \quad (28)$$

It's easy to see that $b_\chi = 1 * \chi$ (Dirichlet convolution), giving us for $\Re(s) > 0$,

$$\sum_{m \geq 1} \frac{b_\chi(m)}{m^{1+s}} = \zeta(1+s) L(1+s, \chi) \quad (29)$$

Insert (29) into (28):

$$-\log H_{1,\chi}(e^{-t}) = \frac{1}{2\pi i} \int_{(c)} \Gamma(s) \zeta(1+s) L(1+s, \chi) t^{-s} ds \quad (30)$$

however, I am unsure of how to continue from here due to the presence of the Dirichlet character and how it may relate to poles.

6 Congruences and eventual periodicity modulo p

In this section we note a nice congruence for the factorially normalized coefficients

$$\phi_{k,\chi}(n) := (n!)^s h_{k,\chi}(n)$$

and deduce eventual periodicity modulo primes. For $k = 1$ this relates to the congruence of Theorem 4.2. For $k \leq 0$ the same mechanism appears to force much stronger p -divisibility, leading to a conjecture.

6.1 A finite recursion modulo p

Theorem 6.1. *Let χ be integer-valued and let $k \leq 1$. Put $s \geq 1$ and write $\phi(n) := \phi_{k,\chi}(n)$. Then for every prime p and every $n \geq 1$,*

$$n \phi(n) \equiv - \sum_{j=1}^{\min(n,p-1)} \sigma_{k-1,\chi}(j) (n)_j^s \phi(n-j) \pmod{p} \quad (31)$$

where $(n)_j := n(n-1) \cdots (n-j+1)$ is the falling factorial and $\sigma_{k-1,\chi}(j) := \sum_{d|j} \chi(d) d^{k-1}$.

Proof. Start from the universal recursion (Proposition 2.1)

$$n h_{k,\chi}(n) = - \sum_{j=1}^n \sigma_{k-1,\chi}(j) h_{k,\chi}(n-j) \quad (n \geq 1)$$

with $h_{k,\chi}(0) = 1$. Multiply both sides by $(n!)^s$ to obtain

$$n (n!)^s h_{k,\chi}(n) = - \sum_{j=1}^n \sigma_{k-1,\chi}(j) (n!)^s h_{k,\chi}(n-j)$$

Using $h_{k,\chi}(n-j) = \phi(n-j)/((n-j)!)^s$, we rewrite the RHS as

$$(n!)^s h_{k,\chi}(n-j) = \left(\frac{n!}{(n-j)!} \right)^s \phi(n-j) = ((n)_j)^s \phi(n-j)$$

and therefore obtain the integral recursion

$$n \phi(n) = - \sum_{j=1}^n \sigma_{k-1,\chi}(j) (n)_j^s \phi(n-j) \quad (32)$$

Now reduce (32) modulo p . If $j \geq p$ then $(n)_j$ is a product of j consecutive integers, hence contains a multiple of p , so $(n)_j \equiv 0 \pmod{p}$ and therefore $(n)_j^s \equiv 0 \pmod{p}$. Thus all terms with $j \geq p$ vanish modulo p , and we obtain (31). Finally, for each fixed $1 \leq j \leq p-1$, the value of $(n)_j \pmod{p}$ depends only on $n \pmod{p}$, so the recursion coefficients in (31) are periodic in n with period p . $\square$

6.2 Eventual periodicity modulo p

Remark 6.2. For $k = 1$ the congruence of Theorem 4.2 gives a much sharper description: for every prime p ,

$$\phi_{1,\chi}(n+p) \equiv \chi(p) \phi_{1,\chi}(n) \pmod{p}$$

so the period divides $2p$ (or p if $\chi(p) = 1$), and if $p \mid \text{cond}(\chi)$ then

$$\phi_{1,\chi}(n) \equiv 0 \pmod{p}$$

for all $n \geq p$.

6.3 A stronger p -divisibility phenomenon for $k \leq 0$

Computations suggest that the eventual periodicity in is often degenerate (period 1) for $k \leq 0$, in the sense that the sequence becomes identically 0 modulo p beyond a sharp cutoff.

Conjecture 6.3. p for $k \leq 0$. Let χ be integer-valued and let $k \leq 0$ (equivalently, $s = 2 - k \geq 2$). Let p be a prime with $p \nmid \text{cond}(\chi)$. Then

$$\phi_{k,\chi}(n) \equiv 0 \pmod{p} \quad \text{for all } n \geq p^2 \quad (33)$$

More strongly, one expects $v_p(\phi_{k,\chi}(n)) \rightarrow \infty$ as $n \rightarrow \infty$

Remark 6.4. The recursion (31) explains why congruence patterns are governed by the falling factorial factors $(n)_j^s$. For $s \geq 2$ these factors carry substantially more p -adic divisibility on average than in the $s = 1$ case, and this may force successive values to acquire an unavoidable factor of p , leading to the cutoff in Conjecture 6.3. A proof would likely combine (32) with systematic lower bounds for $v_p((n)_j^s)$ across the range $1 \leq j \leq p-1$

7 Explicit expansions for conductor 3

Let $\chi = \chi_{-3}$ be the primitive quadratic character modulo 3. $\chi(n) = 0$ if $3 \mid n$, $\chi(n) = 1$ if $n \equiv 1 \pmod{3}$, and $\chi(n) = -1$ if $n \equiv 2 \pmod{3}$. For each $k \in \mathbb{Z}$ we write

$$H_k(q) := H_{k,\chi_{-3}}(q) = \sum_{n \geq 0} h_k(n) q^n, \quad f_k(q) := \sum_{n \geq 0} \phi_k(n) q^n$$

$$\phi_k(n) = (n!)^{\max(0, 2-k)} h_k(n)$$

The coefficients below were computed using the recursion (6).

7.1 Expansions for $-5 \leq k \leq 12$ (for $k \geq 2$, $f_k(q) = H_k(q)$)

$$\begin{aligned}
H_{-5}(q) &= 1 - q + \frac{1}{128}q^2 - \frac{1}{128}q^3 + \frac{127}{32768}q^4 - \frac{9889107}{256000000}q^5 + \frac{859008821}{32768000000}q^6 - \frac{711185585314851}{2698585702400000}q^7 + \dots \\
f_{-5}(q) &= 1 - q + q^2 - 2187q^3 + 17775936q^4 - 1384158528576q^5 + 262951203940451136q^6 - 217701030 \dots \\
H_{-4}(q) &= 1 - q + \frac{1}{64}q^2 - \frac{1}{64}q^3 + \frac{63}{8192}q^4 - \frac{123461}{3200000}q^5 + \frac{38431}{2560000}q^6 - \frac{219359593}{8437500000}q^7 + \frac{433078781}{18000000000}q^8 + \dots \\
f_{-4}(q) &= 1 - q + q^2 - 729q^3 + 1469664q^4 - 9474126848q^5 + 175055105755136q^6 - 1270489542516206 \dots \\
H_{-3}(q) &= 1 - q + \frac{1}{32}q^2 - \frac{1}{32}q^3 + \frac{31}{2048}q^4 - \frac{3781}{200000}q^5 + \frac{1567}{160000}q^6 - \frac{6557691}{525000000}q^7 + \frac{17101729}{1120000000}q^8 - \dots \\
f_{-3}(q) &= 1 - q + q^2 - 243q^3 + 120528q^4 - 294132192q^5 + 15619140133888q^6 - 35541334360508976q^7 + \dots \\
H_{-2}(q) &= 1 - q + \frac{1}{16}q^2 - \frac{1}{16}q^3 + \frac{15}{512}q^4 - \frac{405}{8000}q^5 + \frac{435}{16000}q^6 - \frac{244989}{2800000}q^7 + \frac{11475}{3584}q^8 - \frac{340695}{89600}q^9 + O(q^{10}) \\
f_{-2}(q) &= 1 - q + q^2 - 81q^3 + 9720q^4 - 5743224q^5 + 5638943304q^6 - 14775703356744q^7 + 4218735481 \dots \\
H_{-1}(q) &= 1 - q + \frac{1}{8}q^2 - \frac{1}{8}q^3 + \frac{7}{128}q^4 - \frac{747}{16000}q^5 + \frac{5101}{128000}q^6 - \frac{2184971}{43904000}q^7 + \frac{47752697}{1404928000}q^8 - \frac{44958}{140492}q^9 + O(q^{10}) \\
f_{-1}(q) &= 1 - q + q^2 - 27q^3 + 756q^4 - 80676q^5 + 14874516q^6 - 6371375436q^7 + 2227949831232q^8 - 15 \dots \\
H_0(q) &= 1 - q + \frac{1}{4}q^2 - \frac{1}{4}q^3 + \frac{3}{32}q^4 - \frac{43}{800}q^5 + \frac{887}{16000}q^6 - \frac{8783}{112000}q^7 + \frac{2933}{179200}q^8 - \frac{217039}{4928000}q^9 + O(q^{10}) \\
f_0(q) &= 1 - q + q^2 - 9q^3 + 54q^4 - 774q^5 + 31914q^6 - 2844234q^7 + 133609176q^8 - 9629815896q^9 + O(q^{10}) \\
H_1(q) &= 1 - q + \frac{1}{2}q^2 - \frac{1}{2}q^3 + \frac{1}{8}q^4 + \frac{3}{40}q^5 - \frac{1}{80}q^6 - \frac{129}{560}q^7 + \frac{1137}{4480}q^8 - \frac{1153}{4480}q^9 + O(q^{10}), \\
f_1(q) &= 1 - q + q^2 - 3q^3 + 3q^4 + 9q^5 - 9q^6 - 1161q^7 + 10233q^8 - 93393q^9 + O(q^{10}), \\
H_2(q) &= f_2(q) = 1 - q + q^2 - q^3 + q^5 - q^6 + q^8 - 2q^9 + O(q^{10}), \\
H_3(q) &= f_3(q) = 1 - q + 2q^2 - 2q^3 - q^4 + 6q^5 - 9q^6 + 7q^7 + 4q^8 - 26q^9 + O(q^{10}), \\
H_4(q) &= f_4(q) = 1 - q + 4q^2 - 4q^3 - 6q^4 + 31q^5 - 69q^6 + 95q^7 + 8q^8 - 405q^9 + O(q^{10}), \\
H_5(q) &= f_5(q) = 1 - q + 8q^2 - 8q^3 - 28q^4 + 153q^5 - 517q^6 + 1049q^7 - 103q^8 - 6798q^9 + O(q^{10}), \\
H_6(q) &= f_6(q) = 1 - q + 16q^2 - 16q^3 - 120q^4 + 745q^5 - 3905q^6 + 10879q^7 - 1803q^8 - 119212q^9 + O(q^{10}), \\
H_7(q) &= f_7(q) = 1 - q + 32q^2 - 32q^3 - 496q^4 + 3621q^5 - 29909q^6 + 109977q^7 - 14961q^8 - 2156056q^9 + O(q^{10}), \\
H_8(q) &= f_8(q) = 1 - q + 64q^2 - 64q^3 - 2016q^4 + 17641q^5 - 232009q^6 + 1098735q^7 + 13153q^8 - 39925040q^9 + O(q^{10}), \\
H_9(q) &= f_9(q) = 1 - q + 128q^2 - 128q^3 - 8128q^4 + 86253q^5 - 1817517q^6 + 10915849q^7 + 3580567q^8 - 753170 \dots \\
H_{10}(q) &= f_{10}(q) = 1 - q + 256q^2 - 256q^3 - 32640q^4 + 423265q^5 - 14338785q^6 + 108183359q^7 + 97302017q^8 - 1 \dots \\
H_{11}(q) &= f_{11}(q) = 1 - q + 512q^2 - 512q^3 - 130816q^4 + 2083941q^5 - 113669989q^6 + 1071363257q^7 + 200431763 \dots \\
H_{12}(q) &= f_{12}(q) = 1 - q + 1024q^2 - 1024q^3 - 523776q^4 + 10289401q^5 - 904025849q^6 + 10611784975q^7 + 36900 \dots
\end{aligned}$$

8 Further questions and directions

We list a few directions that naturally follow from the results above.

- **Minimal denominators** Determine the minimal integer $D_{k,\chi}(n)$ such that $D_{k,\chi}(n) h_{k,\chi}(n) \in \mathbb{Z}$; compare to the uniform bound $(n!)^{\max(0, 2-k)}$. This should be very interesting, and yield congruences similar to those with ϕ but with more structure unique to H (perhaps)
- **Congruences and periodicities** For fixed (k, χ) , find more congruences for $\phi_{k,\chi}(n)$
- **Divisors** Looking purely at $h_{k,\chi}$, what are the divisibility properties, prime divisors, etc of the coefficients. One thing I was taking a look at was [7], at the Dieudonné–Dwork criterion.

- **Asymptotics** Finish (30) to get a closed form expression. Study the asymptotics

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